Euclid's Elements — Book I
Proposition 48

Formal Reconstruction — Technical Documentation

Jurek Różański
Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJteamLab

The converse of the Pythagorean theorem

1 Introduction

Proposition I.48 is the converse of the Pythagorean theorem in the synthetic language of Euclid's Book I. Given a nondegenerate triangle ABC, suppose that the square on BC has the same content as the sum of the squares on AB and AC. The conclusion is that the angle BAC is right.

The statement must be read at the same level as Proposition I.47. It is not the numerical assertion

$$BC^2 = AB^2 + AC^2$$

followed by a calculation with real numbers. In the current reconstruction the three squares are explicit geometric objects and the hypothesis is an instance of

```
HilbertScissorsEquicomplementable
```

relating their formal scissors terms. The conclusion is the synthetic predicate

```
HilbertRightAngle Geo B A C
```

with no degree measure attached to the angle.

Classically, Euclid's proof is short. Construct a second triangle ADC with $AD \cong AB$ and $\angle DAC$ right. Proposition I.47 gives

$$\mathrm{Sq}(DC) \approx \mathrm{Sq}(DA) + \mathrm{Sq}(AC),$$

where $\approx$ denotes equal content. Since $AD \cong AB$, the square on DA equals the square on BA. The hypothesis then gives

$$\mathrm{Sq}(DC) \approx \mathrm{Sq}(BC).$$

Euclid passes immediately from equality of these squares to equality of their sides, so $DC \cong BC$. SSS applied to DAC and BAC then transfers the right angle at A.

The formal reconstruction exposes that the sentence

equal squares have equal sides

is the difficult direction. The established scissors calculus gives many ways to prove equal content from geometry, but equal content does not by itself contain an order relation that can distinguish a figure from a proper part. Proposition I.48 therefore needs a strictness argument.

The final file proves this strictness rather than postulating an I.48-specific cancellation theorem. A strict inequality between two square sides is converted into an inner square inside the larger square. The larger square is then dissected into the inner square plus a nonempty triangulated remainder. The project-wide scissors positivity principle shows that the remainder has nonzero content, so the two squares cannot be equicomplementable. Segment trichotomy leaves congruence as the only possibility for sides of equicomplementable squares.

Thus the final proof has two quite different scales:

1. a short Euclidean outer proof using a constructed right triangle, I.47, SSS, and right-angle transport;
2. a longer converse-content core proving that equal square content forces congruent sides.

There is no proposition-local axiom in the final source and no `sorry` or `admit`. The complete project build is clean. The proof does, however, depend transitively on the project-wide temporary content assumption `hilbert_scissors_triangle_positive`. This status is recorded explicitly below.

2 The Classical Configuration

Let ABC be the given nondegenerate triangle. The formal theorem receives three explicit squares as data:

$$BCPQ, \qquad ABRS, \qquad ACTU,$$

with the hypothesis

$$BCPQ \approx ABRS + ACTU.$$

Here and throughout this document $\approx$ denotes the relation

```
HilbertScissorsEquicomplementable
```

not equality of numerical areas.

At A the proof constructs a new point D such that

$$AD \cong AB, \qquad \angle DAC \text{ is right.}$$

This is the formal counterpart of Euclid's use of I.11 followed by I.3. The point D need not be placed in the diagram by assuming the conclusion about $\angle BAC$. In particular, the Lean theorem does not require D, A, B to be collinear. The construction only needs the prescribed perpendicular and the prescribed length.

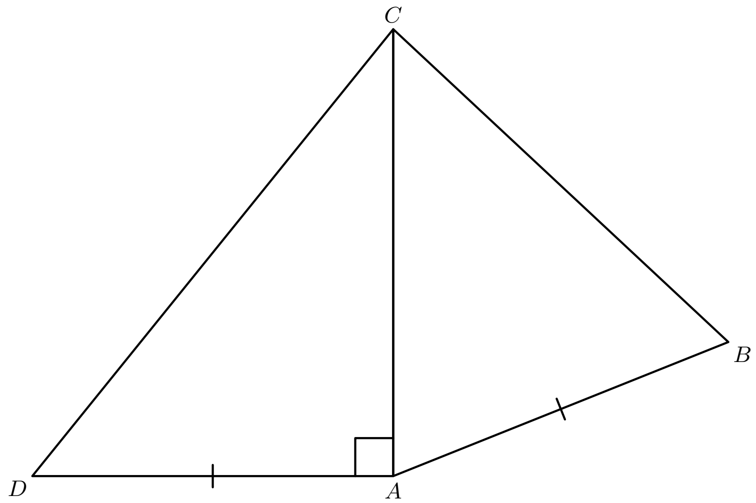

Figure 1: Euclid's classical comparison configuration. The auxiliary segment AD is erected at right angles to AC with $AD \cong AB$, and DC is joined. The displayed placement of B and D on opposite sides of AC is the classical diagrammatic arrangement; it is not an additional side hypothesis of the Lean theorem.

The triangle ADC is therefore a genuine right triangle. Applying I.47 to it constructs three further squares: one on DC, one on AD, and one on AC, together with the equal-content relation

$$\mathrm{Sq}(DC) \approx \mathrm{Sq}(AD) + \mathrm{Sq}(AC).$$

The two leg squares produced by I.47 are not definitionally the same objects as the given squares on AB and AC. A square is not determined in the formal language by its side alone. The theorem `i48_square_transport` therefore transports content between squares erected on congruent segments.

After the two transports, transitivity gives

$$\mathrm{Sq}(DC) \approx \mathrm{Sq}(AB) + \mathrm{Sq}(AC) \approx \mathrm{Sq}(BC).$$

The theorem `i48_congruent_of_equal_squares` converts the last content equality into

$$DC \cong BC.$$

Now the triangles DAC and BAC have three corresponding congruent sides:

$$DA \cong BA, \qquad AC \cong AC, \qquad DC \cong BC.$$

SSS gives the required angle congruence at A, and rightness is transported from $\angle DAC$ to $\angle BAC$.

3 Classical Proof

Euclid's proof can be represented by a very small dependency graph.

First erect AD at right angles to AC and choose it so that

$$AD \cong AB.$$

This is the I.11/I.3 construction stage. Join DC. Since $\angle DAC$ is right, I.47 gives

$$\mathrm{Sq}(DC) \approx \mathrm{Sq}(DA) + \mathrm{Sq}(AC).$$

The equality $DA \cong AB$ allows the square on DA to be replaced by the square on BA. Therefore

$$\mathrm{Sq}(DC) \approx \mathrm{Sq}(BA) + \mathrm{Sq}(AC).$$

By the hypothesis of I.48, the right-hand side is also equal to the square on BC. Hence

$$\mathrm{Sq}(DC) \approx \mathrm{Sq}(BC).$$

Euclid then concludes that

$$DC \cong BC.$$

Finally, with

$$DA \cong BA, \qquad AC \cong AC, \qquad DC \cong BC,$$

Proposition I.8 gives

$$\angle DAC \cong \angle BAC.$$

Since $\angle DAC$ is right, $\angle BAC$ is right.

The classical DAG is therefore

$$\begin{array}{c}
\text{I.11 + I.3: construct } AD \perp AC,\ AD \cong AB \\
\Downarrow \\
\text{I.47 on } ADC \\
\Downarrow \\
\mathrm{Sq}(DC) \approx \mathrm{Sq}(DA) + \mathrm{Sq}(AC) \\
\Downarrow \\
AD \cong AB \quad + \quad \text{I.48 hypothesis} \\
\Downarrow \\
\mathrm{Sq}(DC) \approx \mathrm{Sq}(BC) \\
\Downarrow \\
DC \cong BC \\
\Downarrow \\
\text{I.8 / SSS} \\
\Downarrow \\
\angle BAC \text{ right.}
\end{array}$$

The formal reconstruction follows this outer structure closely, but the implication

$$\mathrm{Sq}(DC) \approx \mathrm{Sq}(BC) \Longrightarrow DC \cong BC$$

expands into the main new proof of Proposition I.48.

4 Converse Content, Proper Parts, and Positivity

4.1 Why the converse direction is different

The forward content theorems of I.35–I.47 start with geometric information and produce content equality. Congruence, parallelism, common bases, triangulation, and finite addition are enough for this direction.

I.48 reverses the direction at a decisive point. It starts with equality of content of two squares and must recover a metric statement about their sides. For this implication, an equivalence relation and an additive scissors calculus are not enough. One needs a strictness principle saying that adjoining a genuine nondegenerate piece changes content.

The current project isolates that strictness in the global declaration

```
hilbert_scissors_triangle_positive
```

from `HilbertScissorsPositivity`. In schematic form it says

$$P + T \not\approx P$$

whenever T is a nondegenerate triangle term. It is configuration-free: it does not mention squares, betweenness, or Proposition I.48.

The local I.48 proof derives the needed square monotonicity from this one principle. It does not assume a new axiom saying that equal squares have equal sides.

4.2 From a shorter side to an inner square

Suppose two squares have sides AB and $A'B'$ and assume

$$AB < A'B'.$$

The relation `HilbertSegmentLess` contains a proper cut on the longer segment. Thus a point X is obtained with

$$A' - X - B', \qquad A'X \cong AB.$$

Since adjacent sides of the large square are congruent, the strict comparison is transported to $A'D'$, giving another point Y with

$$A' - Y - D', \qquad A'Y \cong AB.$$

Hence

$$A'X \cong A'Y.$$

The rays $A'X$ and $A'B'$ coincide, as do $A'Y$ and $A'D'$. The right angle of the large square is therefore transported to $\angle YA'X$. Completing the equal perpendicular sides constructs a square

$$A'XZY$$

strictly contained at the corner A' of the larger square. The theorem `i48_square_transport` simultaneously shows that the original smaller square has the same content as this inner square.

This stage is packaged by

```
i48_aux_square_side_cuts
i48_aux_complete_square
i48_aux_inner_square_of_less
i48_aux_inner_square_transport
```

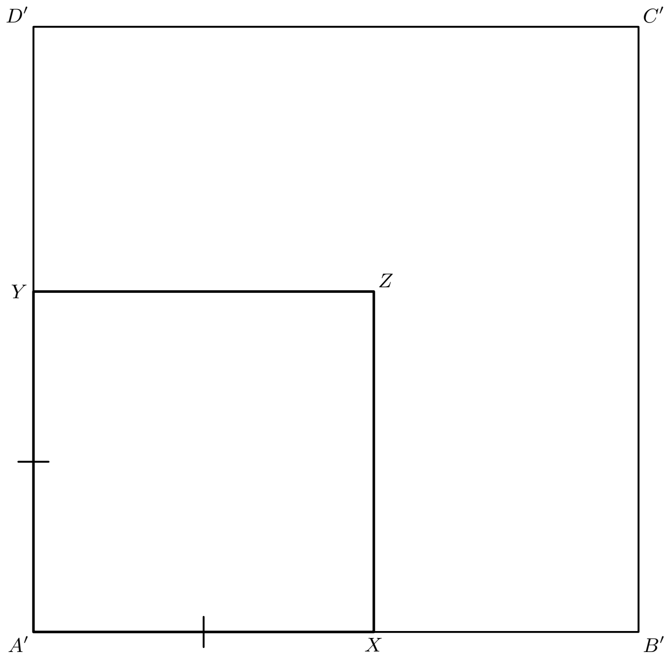

Figure 2: The geometric state produced from $AB < A'B'$. The strict cuts $A' - X - B'$ and $A' - Y - D'$ determine equal initial segments on adjacent sides of the larger square, and the construction completes them to the inner square $A'XZY$. No remainder cuts have yet been introduced.

4.3 The exact remainder decomposition

The next stage is an exact scissors calculation, not merely an intuitive claim that the inner square is a proper subset.

For any parallelogram $ABCD$, the theorem

```
i48_aux_parallelogram_double_corner
```

proves

$$ABCD \sim ABD + ABD,$$

where $\sim$ denotes `HilbertScissorsEq`. The proof uses the two triangulations of a parallelogram and I.34 to identify the two diagonal halves.

Now assume

$$A - X - B, \qquad A - Y - D.$$

Successive triangle splits give

$$ABD \sim AXY + YDX + XBD.$$

This is the theorem

```
i48_aux_corner_triangle_split
```

Combining the two doubled-triangle descriptions of the outer and inner squares gives the exact identity

$$\begin{aligned}\mathrm{Sq}(ABCD) \sim\ & \mathrm{Sq}(AXZY) \\ & + (YDX + XBD) + (YDX + XBD).\end{aligned}$$

This is `i48_aux_square_split_with_remainder`.

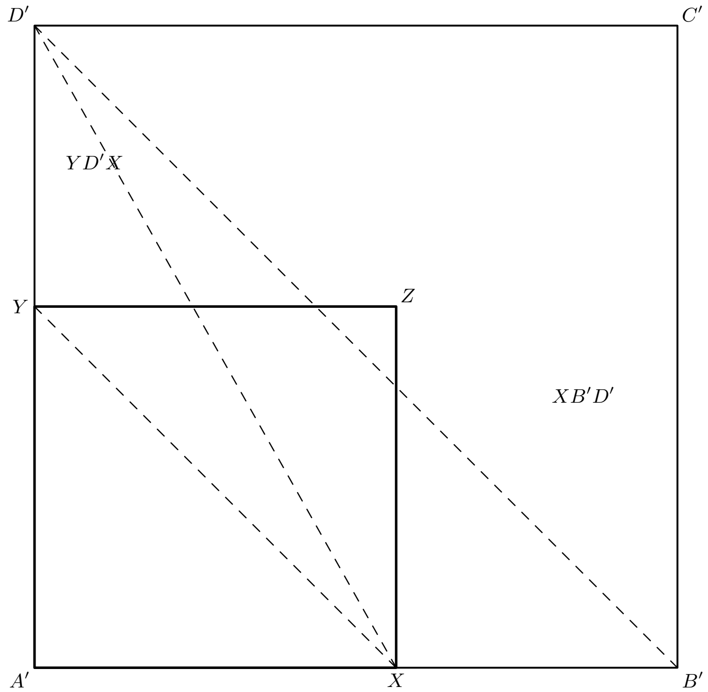

Figure 3: The additional cuts used by the exact scissors decomposition. The diagonal $B'D'$, the inner diagonal XY, and the segment $D'X$ split one diagonal half into the inner triangle and the two remainder triangles $YD'X$ and $XB'D'$. Doubling this half produces the four triangle remainder in `i48_aux_square_split_with_remainder`.

The remainder is therefore not an abstract difference. It is an explicit formal sum of four triangles. This is important because the current scissors language has no primitive subtraction of figures and no numerical area.

4.4 Why the remainder is genuinely nonzero

The two triangle types occurring in the remainder are

$$YDX, \qquad XBD.$$

The theorem

```
i48_aux_remainder_triangles_nondegenerate
```

proves that both are noncollinear from the outer square and the strict betweenness relations. The four pieces are then packaged as the nonempty triangulated figure

$$[YDX,\ XBD,\ YDX,\ XBD].$$

The theorem `i48_aux_remainder_figure` verifies both nonemptiness and nondegeneracy of every member of this list and identifies its `rectilinealTerm` with the exact remainder above.

To apply the global triangle positivity axiom to an arbitrary nonempty triangulated remainder, the proof constructs a reduction chain. Proposition I.45 converts the rectilineal figure into an equicomplementable parallelogram. The local theorem

```
i48_aux_parallelogram_to_triangle
```

then converts that parallelogram into one nondegenerate triangle. Hence every such remainder is equicomplementable with a nondegenerate triangle. If the remainder were equicomplementable with zero, transitivity would make that triangle equicomplementable with zero, contradicting `hilbert_scissors_triangle_positive`.

The corresponding chain is

$$\begin{array}{c} \text{nonempty triangulated nondegenerate remainder} \\ \Downarrow \quad \text{I.45} \\ \text{equicomplementable parallelogram} \\ \Downarrow \\ \text{equicomplementable nondegenerate triangle} \\ \Downarrow \quad \text{positivity} \\ \text{remainder} \not\approx 0. \end{array}$$

This is implemented by

```
i48_aux_parallelogram_to_triangle
i48_aux_rectilineal_to_triangle
i48_aux_rectilineal_positive
```

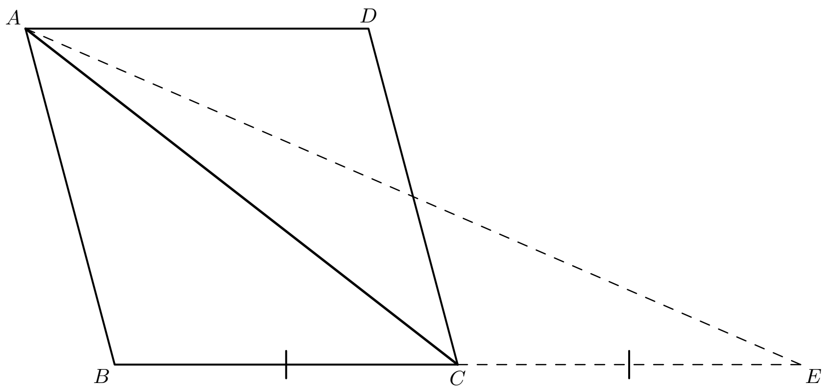

Figure 4: The geometric core of `i48_aux_parallelogram_to_triangle`. The side BC is extended to E with $B - C - E$ and $BC \cong CE$, so C is the midpoint of BE. The diagonal AC divides the parallelogram by I.34 and is simultaneously the median used by I.42 in triangle ABE. This converts the parallelogram content to the content of one nondegenerate triangle.

4.5 Cancellation and the proper-part theorem

Suppose

$$Q \sim P + R$$

and $R \not\approx 0$. If one also had $Q \approx P$, transitivity would give

$$P + R \approx P.$$

The local algebraic cancellation theorem

```
i48_aux_equicomplementable_cancel_right
```

removes the common term P after commuting the formal multiset sum, yielding $R \approx 0$, a contradiction.

Therefore

$$Q \sim P + R, \qquad R \not\approx 0 \quad \Longrightarrow \quad Q \not\approx P.$$

The generic packaging is

```
i48_aux_not_equal_of_positive_remainder
i48_aux_not_equal_of_rectilineal_remainder
```

Applied to the inner-square decomposition, it gives

```
i48_aux_inner_square_proper_part
```

and therefore

$$AB < A'B' \quad \Longrightarrow \quad \mathrm{Sq}(AB) \not\approx \mathrm{Sq}(A'B').$$

The last statement is

```
i48_aux_square_not_equal_of_less
```

4.6 Segment trichotomy completes the converse

Now suppose two squares are equicomplementable. The strict inequality $AB < A'B'$ is impossible by the theorem just proved. By symmetry the reverse strict inequality $A'B' < AB$ is also impossible. Both sides are nondegenerate because they are sides of squares. The Book Zero segment trichotomy theorem therefore leaves only

$$AB \cong A'B'.$$

This is the content of

```
i48_congruent_of_equal_squares
```

The theorem is the formal replacement for Euclid's apparently immediate step from equal squares to equal sides.

5 Proof Architecture of the Reconstruction

The final Lean source is best read as eight mathematical stages.

5.1 Step 1: construct the prescribed perpendicular segment

The theorem

```
i48_erect_perpendicular
```

packages the I.11/I.3 operation needed by Euclid. In ASCII notation its interface is

```
theorem i48_erect_perpendicular
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A C P Q : Geo.Point)
    (hAC : Not (A = C)) :
    exists D : Geo.Point,
      Not (Collinear Geo D A C) /\
      HilbertRightAngle Geo D A C /\
      Geo.Congruent A D P Q
```

The proof first extends CA beyond A so that the established right-angle existence theorem can be applied at A. It then lays off the independently prescribed segment PQ on the resulting perpendicular ray. Same-ray angle transport shows that replacing the initially constructed point by the laid-off point preserves the right angle. Finally the angle arms are swapped into the orientation DAC used by I.48.

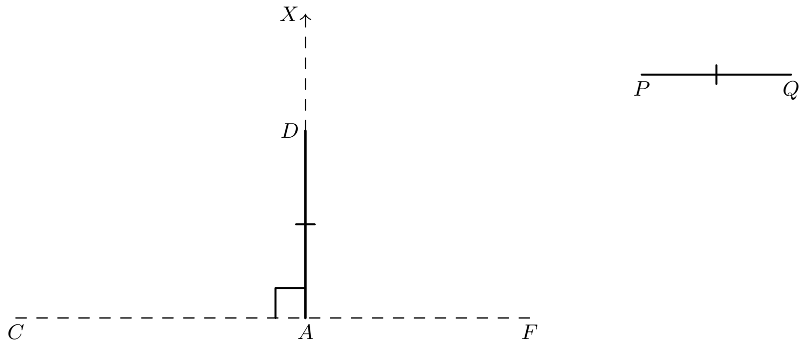

Figure 5: The hidden witnesses in `i48_erect_perpendicular`. First CA is extended beyond A to F; I.11 supplies a perpendicular point X; then the prescribed segment PQ is laid off as AD on the same ray from A. The drawing chooses one order of D and X only for visibility: the proof uses `SameRay`, not a betweenness relation between those two points.

This helper is mathematically reusable: it erects a perpendicular at a specified endpoint with a length unrelated to the carrier segment.

5.2 Step 2: transport content between squares on congruent sides

The theorem

```
i48_square_transport
```

proves that squares erected on congruent segments are equicomplementable. The proof is genuinely geometric rather than definitional.

First the congruence of one pair of sides is propagated around the two squares. The right angles at corresponding vertices are congruent by `hilbert_all_right_angles_congruent`. SAS then identifies the corresponding diagonals. Each of the two diagonal halves is matched by SSS, and the two triangle congruences are added as a `HilbertScissorsEq`. Finally exact scissors equality is weakened to equicomplementability.

This theorem formalizes Euclid's sentence that equal straight lines have equal squares described on them. It is used twice in the public I.48 proof: for the square on AD versus the given square on AB, and for the square on AC produced by I.47 versus the given square on AC.

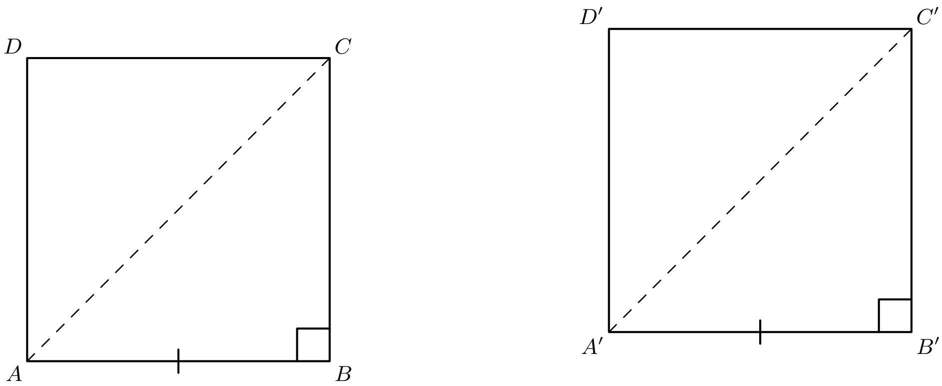

Figure 6: The geometric content of `i48_square_transport`. Congruent base sides propagate around the two squares; right angles at B and B' give the congruent diagonals AC and $A'C'$ by SAS, after which the two diagonal halves correspond by SSS. The figure records the triangulation, not a numerical area argument.

5.3 Step 3: turn strict side inequality into an inner square

The helper

```
i48_aux_square_side_cuts
```

uses `HilbertSegmentLess` together with the side congruences of I.46 to cut equal proper initial segments on two adjacent sides of the larger square.

The theorem

```
i48_aux_complete_square
```

completes equal perpendicular segments to a square. It first constructs the fourth vertex of a parallelogram, then uses I.34, opposite-side congruences, and right-angle transport to establish all four square conditions.

The theorem

```
i48_aux_inner_square_of_less
```

combines these steps. It must prove that the cut rays are noncollinear and that their angle is right. The latter is obtained by transporting the right angle of the large square along the same rays determined by the betweenness data.

Finally

```
i48_aux_inner_square_transport
```

uses `i48_square_transport` to identify the content of the original smaller square with the newly constructed inner square.

5.4 Step 4: expose the remainder by exact triangle splits

The theorem

```
i48_aux_parallelogram_double_corner
```

retriangulates a parallelogram along the diagonal joining the other two vertices. I.34 identifies the two resulting triangles, so the parallelogram term is scissors-equal to two copies of a corner triangle.

Next,

```
i48_aux_corner_triangle_split
```

uses the two strict betweenness relations to split that corner triangle into one inner-square triangle and two remainder triangles.

The theorem

```
i48_aux_square_split_with_remainder
```

performs the same substitution in both copies and then replaces the two inner triangles by the inner square term. The output is an exact `HilbertScissorsEq`, not merely an equicomplementability statement.

5.5 Step 5: certify positivity of the explicit remainder

The theorem

```
i48_aux_remainder_triangles_nondegenerate
```

shows that neither remainder triangle collapses. Both proofs are incidence arguments: a hypothetical collinearity is transported through the betweenness carriers until it contradicts noncollinearity of three vertices of the outer square.

The theorem

```
i48_aux_remainder_figure
```

packages the four pieces as a `HilbertTriangulatedFigure` and proves that the corresponding `rectilinealTerm` is exactly the remainder in the square split.

The two conversion helpers

```
i48_aux_parallelogram_to_triangle
i48_aux_rectilineal_to_triangle
```

then reduce an arbitrary nonempty triangulated nondegenerate figure to one nondegenerate triangle. The second helper calls I.45. The first uses a segment extension, I.42's median-content theorem, I.34, and an exact triangle split.

The result

```
i48_aux_rectilineal_positive
```

is the point at which the project-wide positivity axiom enters I.48.

5.6 Step 6: derive strict monotonicity of square content

The theorem

```
i48_aux_equicomplementable_cancel_right
```

is a purely algebraic consequence of the definition of equicomplementability. It is not a new positivity assumption. It rearranges the complement witnesses so that a common right summand can be cancelled.

Together with positivity this yields

```
i48_aux_not_equal_of_positive_remainder
i48_aux_not_equal_of_rectilineal_remainder
i48_aux_inner_square_proper_part
i48_aux_square_not_equal_of_less
```

The mathematical conclusion is strict monotonicity:

$$AB < A'B' \Longrightarrow \mathrm{Sq}(AB) \not\approx \mathrm{Sq}(A'B').$$

5.7 Step 7: recover side congruence from equal square content

The central converse theorem has the current source signature

```
theorem i48_congruent_of_equal_squares
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D A' B' C' D' : Geo.Point)
    (hSquare : IsSquare Geo A B C D)
    (hSquare' : IsSquare Geo A' B' C' D')
    (hEqual :
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo A B C D)
        (hilbertParallelogramTerm Geo A' B' C' D')) :
    Geo.Congruent A B A' B'
```

The proof excludes both strict segment orders using `i48_aux_square_not_equal_of_less`. It then applies `bookZero_31_trichotomy1`. This is the exact point where the strict-content work reconnects to ordinary Hilbert segment geometry.

5.8 Step 8: the public I.48 theorem

The public theorem now becomes short. Construct D, apply I.47 to ADC, transport the two leg squares, compose with the given content equality, use `i48_congruent_of_equal_squares` to obtain $DC \cong BC$, apply SSS to DAC and BAC, and transport the right angle.

The final proof therefore has the schematic form

$$\begin{array}{c} AD \perp AC,\ AD \cong AB \\ \Downarrow \\ \text{I.47 on } ADC \\ \Downarrow \\ \mathrm{Sq}(DC) \approx \mathrm{Sq}(AD) + \mathrm{Sq}(AC) \\ \Downarrow \quad \text{square transport} \\ \mathrm{Sq}(DC) \approx \mathrm{Sq}(AB) + \mathrm{Sq}(AC) \\ \Downarrow \quad \text{I.48 hypothesis} \\ \mathrm{Sq}(DC) \approx \mathrm{Sq}(BC) \\ \Downarrow \quad \text{strict square monotonicity + trichotomy} \\ DC \cong BC \\ \Downarrow \quad \text{SSS} \\ \angle DAC \cong \angle BAC \\ \Downarrow \\ \angle BAC \text{ right.} \end{array}$$

Diagrammatic audit. Figure 1 is retained as the classical Euclidean configuration. Figures 2, 3, 4, 5, and 6 record the distinct formal geometric states introduced by the current Lean proof: the strict inner square, the explicit remainder cuts, the parallelogram-to-triangle positivity bridge, the hidden perpendicular-construction witnesses, and diagonal transport between squares. No separate figures are added for congruence symmetry, transitivity, scissors cancellation, segment trichotomy, or the final SSS/right-angle transport, since those steps do not introduce a new underlying geometric configuration.

6 Lean Formalization

6.1 Imports

The current source imports exactly

```
import CGJteamLab.Proposition47
import CGJteamLab.Proposition42
import CGJteamLab.Proposition45
import CGJteamLab.HilbertScissorsPositivity
```

These imports reveal the real formal structure. I.47 supplies the forward Pythagorean content equality. I.42 and I.45 are used inside the proof that a nonempty rectilineal remainder has positive content. The positivity module supplies the one project-wide strict-content assumption.

6.2 The public proposition

In ASCII-normalized documentation notation, the public theorem is

```
theorem euclid_proposition_48
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C P Q R S T U : Geo.Point)
    (hABC : Not (Collinear Geo A B C))
```

```
    (hSqBC : IsSquare Geo B C P Q)
    (hSqAB : IsSquare Geo A B R S)
    (hSqAC : IsSquare Geo A C T U)
    (hEqual :
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo B C P Q)
        (hilbertParallelogramTerm Geo A B R S +
         hilbertParallelogramTerm Geo A C T U)) :
    HilbertRightAngle Geo B A C
```

The three squares are hypotheses rather than existentially reconstructed inside I.48. This is deliberate. The content hypothesis refers to the actual square terms, and the current geometry language does not quotient all squares on congruent sides into one definitional object. Equality between different choices is instead proved by `i48_square_transport`.

6.3 The strict square inequality theorem

The key monotonicity theorem has the form

```
theorem i48_aux_square_not_equal_of_less
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D A' B' C' D' : Geo.Point)
    (hSquare : IsSquare Geo A B C D)
    (hSquare' : IsSquare Geo A' B' C' D')
    (hLess : HilbertSegmentLess Geo A B A' B') :
    Not
      (HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo A B C D)
        (hilbertParallelogramTerm Geo A' B' C' D'))
```

The statement contains no numerical length and no numerical area. Strict segment comparison is entirely synthetic, and the conclusion is strict inequality at the level of equicomplementability.

6.4 The derived square-side converse

The theorem consumed by the public proposition is

```
theorem i48_congruent_of_equal_squares
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D A' B' C' D' : Geo.Point)
    (hSquare : IsSquare Geo A B C D)
    (hSquare' : IsSquare Geo A' B' C' D')
    (hEqual :
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo A B C D)
        (hilbertParallelogramTerm Geo A' B' C' D')) :
    Geo.Congruent A B A' B'
```

This theorem was previously the unresolved local assumption in the I.48 sketch. In the current source it is a theorem proved by the full strictness chain described above.

6.5 Right-angle congruence

The other former local assumption has also disappeared. The wrapper theorem

```
i48_right_angles_congruent
```

is now an immediate call to

```
hilbert_all_right_angles_congruent
```

Thus Euclid's Postulate 4 is supplied by the established Hilbert right-angle infrastructure rather than by a proposition-local axiom.

7 Relationship with Earlier Propositions

7.1 Propositions I.3 and I.11

Euclid first erects a perpendicular at A and then makes its new segment equal to AB. The formal proof packages these operations in `i48_erect_perpendicular`. Segment extension, right-angle existence, and Hilbert segment construction perform the work that the classical proof cites as I.11 and I.3.

The exact Book I theorem names need not appear in the call graph because the corresponding construction operations already live in the developed Hilbert interface. The classical and formal dependency graphs are therefore not identical.

7.2 Proposition I.8

The last geometric step is SSS. Once $DC \cong BC$ has been recovered, the triangles DAC and BAC have all three corresponding sides congruent. The Lean proof calls `HilbertSSS` and then uses the returned angle congruence to transport rightness.

Thus the classical role of I.8 is present exactly, although the formal source uses the Hilbert theorem rather than the Book I proposition wrapper.

7.3 Proposition I.34

I.34 enters the strictness core rather than the final Euclidean shell. It is used to compare diagonal halves of parallelograms. This supports both the "parallelogram equals two copies of a corner triangle" lemma and the parallelogram-to-triangle reduction used in positivity.

7.4 Proposition I.42

The local theorem `i48_aux_parallelogram_to_triangle` uses `i42_median_bisects_area`. A segment is extended beyond a vertex so that a midpoint configuration is created; I.42 then identifies the two half-triangles in content. Together with the diagonal decomposition of the parallelogram this produces one triangle equicomplementable with the whole parallelogram.

I.42 is therefore not part of Euclid's visible I.48 proof. It is a formal support theorem used only to convert a general positive rectilineal remainder to the single-triangle form expected by the global positivity axiom.

7.5 Proposition I.45

I.45 is the main bridge from an arbitrary finite rectilineal remainder to a parallelogram. The theorem `i48_aux_rectilineal_to_triangle` first applies `euclid_proposition_45` to the nonempty triangulated figure, then applies the local parallelogram-to-triangle theorem.

This is a particularly useful example of the difference between the classical and formal DAGs. Euclid never needs to replace the square remainder by an arbitrary equivalent parallelogram in the visible I.48 proof. The formal proof does so only because the global strictness principle is stated for a single nondegenerate triangle.

7.6 Proposition I.46

The square predicate and its side geometry come from the infrastructure developed for I.46. In particular,

```
euclid_proposition_46_side
```

is used when a strict comparison with one side of the large square must be transported to the adjacent side.

I.46 also provides the conceptual background for treating a square as an explicit geometric object rather than as a numerical square of a segment length.

7.7 Proposition I.47

I.47 is the headline dependency of the proposition. It is applied once to the constructed right triangle ADC. Its output consists of three explicit squares and an equicomplementability relation between the hypotenuse square and the sum of the two leg squares.

The public I.48 proof then uses

```
i47_aux_equicomplementable_add
```

to replace both I.47 leg squares simultaneously by the squares occurring in the I.48 hypothesis.

The contrast between the propositions is fundamental:

$$\begin{aligned} &\text{I.47:} \quad \text{right angle} \Longrightarrow \text{equal square content}, \\ &\text{I.48:} \quad \text{equal square content} \Longrightarrow \text{right angle}. \end{aligned}$$

The reverse arrow requires the new strictness argument.

8 Hilbert and Book Zero Components Used

8.1 Segment order and trichotomy

The local definition `HilbertSegmentLess` supplies the witness used to cut the shorter length from the longer square side. Congruence transport on the right endpoint moves that strict comparison to the adjacent side of the square.

At the end of the converse-content argument,

```
bookZero_31_trichotomy1
```

is decisive. After both strict orders are excluded, trichotomy returns segment congruence. No numerical length function is needed.

8.2 Betweenness and SameRay

The strict comparison witnesses produce

$$A' - X - B', \qquad A' - Y - D'.$$

Betweenness provides nondegeneracy and carrier collinearity. SameRay conversion then shows that the rays $A'X$ and $A'B'$ agree and that the rays $A'Y$ and $A'D'$ agree. The right angle of the large square can therefore be transported to the two cut segments.

These order data are central. Without strict betweenness, the constructed inner square could collapse onto the outer boundary and the proper-part argument would fail.

8.3 Parallelogram and square infrastructure

The proof repeatedly extracts the `IsParallelogram` component of `IsSquare`. The main tools include opposite-side congruences, noncollinearity of parallelogram vertices, construction of a fourth vertex, and the two diagonal triangulations.

This infrastructure turns the intuitive phrase "a square is twice one of its diagonal triangles" into an exact theorem in the scissors calculus.

8.4 Congruence, SAS, SSS, and right-angle transport

The square transport theorem uses both SAS and SSS. SAS first establishes congruence of the diagonals of two squares on congruent sides; SSS then matches the two triangle halves. In the public theorem, SSS appears again in the classical role of I.8.

Right angles are handled synthetically through

```
hilbert_all_right_angles_congruent
hilbert_right_angle_transport
```

No angle measure or arithmetic with 90 degrees is introduced.

8.5 Scissors operations

The content proof uses the established constructors and operations

```
HilbertScissorsEq.refl
HilbertScissorsEq.symm
HilbertScissorsEq.trans
HilbertScissorsEq.add
HilbertScissorsEq.split
scissors_congruent
equicomplementable_of_scissorsEq
equicomplementable_add
equicomplementable_symm
equicomplementable_trans
```

The important distinction is that exact dissection is recorded by `HilbertScissorsEq`, while the public equality of contents is expressed by `HilbertScissorsEquicomplementable`. The proof moves from exact splits to equicomplementability only when required.

8.6 Scissors positivity

The only strict-content assumption is inherited from `HilbertScissorsPositivity`:

```
hilbert_scissors_triangle_positive
```

The local I.48 theorems derive all square-specific consequences from it. Consequently there is no active axiom in `Proposition48.lean`, but the public theorem is not independent of the project-wide positivity assumption.

9 Formalization Notes

9.1 Geometry

The geometry layer of I.48 has two distinct parts.

The outer Euclidean proof constructs only one new point D. It proves

- D, A, C are noncollinear;
- $\angle DAC$ is right;
- $AD \cong AB$.

After this, I.47 supplies the squares needed for the comparison.

The strictness core has a different geometry. From a strict side inequality it constructs X and Y as proper points on two adjacent sides of the larger square, proves the cut rays are perpendicular and equal, completes an inner square, and proves the two remainder triangle types are nondegenerate.

No coordinate system is introduced at either level.

9.2 Content

The content layer is the central novelty of the proposition.

Forward transport is routine: congruent squares are shown equicomplementable, I.47 gives a sum of two square contents, and transitivity combines the equalities.

The converse is not routine. The proof needs the implication

$$\text{strictly shorter side} \Longrightarrow \text{strictly smaller square content.}$$

This is derived by an exact proper-part decomposition plus positivity. The proof never introduces a real-valued area, multiplication of segment lengths, or subtraction of area values.

The resulting theorem `i48_congruent_of_equal_squares` is therefore synthetic: it recovers segment congruence from content equality using segment order and a De Zolt-type strictness principle.

9.3 Representation

Several representation issues are mathematically harmless but formally visible.

First, a square is an `IsSquare` predicate on four named points. Two squares on congruent sides are not definitionally identical, so explicit content transport is required.

Second, a parallelogram term has a fixed triangulation in its definition, but the proper-part argument is easiest in the other diagonal triangulation. The theorem

```
i48_aux_parallelogram_double_corner
```

bridges those two representations.

Third, the remainder is a multiset-level sum of repeated triangle terms. The proof uses associativity and commutativity of `Multiset` addition to regroup the two copies of the corner decomposition. This bookkeeping does not add geometric content.

Fourth, I.45 consumes a triangulated figure represented by the type

```
HilbertTriangulatedFigure Geo
```

while the remainder of the square split is a `HilbertScissorsTerm`. The theorem

```
i48_aux_remainder_figure
```

explicitly identifies the list's `rectilinealTerm` with the multiset expression produced by the split.

9.4 Hidden configuration data

The Lean proof makes the following data explicit even though a classical diagram tends to suppress them:

- $A \neq C$ is extracted from noncollinearity before the perpendicular construction can be called;

- the constructed point D must be proved noncollinear with A, C;
- replacing the initial perpendicular point by the point obtained from segment layoff requires SameRay angle transport;
- the two squares on AC occurring in I.47 and in the hypothesis are distinct named objects and must be transported;
- a strict side comparison must be transported from $A'B'$ to $A'D'$;
- the two cut points X, Y must be strict interior points of the relevant sides;
- the cut rays must be proved noncollinear before a right angle can be transported to them;
- both remainder triangle types must be proved nondegenerate;
- the four-triangle remainder must be proved nonempty as a list before I.45 can be applied.

The formal proof does *not* need the classical pictorial convention that the newly erected D lies on the opposite side of AC from B. SSS at the end works from side congruences and the established right angle without such a side-of-line hypothesis.

9.5 Axiomatic status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

so Proposition I.48 is Euclidean in the current project API. Group IV is available through the imported Euclidean plane infrastructure and is used transitively in the square, parallelogram, I.45, and I.47 machinery.

No continuity typeclass is assumed by `Proposition48.lean`. No Archimedean axiom is introduced in the file. No numerical field of segment lengths and no numerical area function are used.

There is no proposition-local `axiom`. The former assumptions "all right angles are congruent" and "equal squares have congruent sides" are both theorems in the final source. The second of these, however, depends on the project-wide temporary content axiom

```
hilbert_scissors_triangle_positive
```

imported from `HilbertScissorsPositivity`. Thus the correct status is: I.48 adds no new foundational debt, but it consumes the already isolated scissors-positivity debt.

10 Source Audit

10.1 Euclid and Wilkins

The classical sources agree on the short proof architecture. A right triangle ADC is constructed with AC common, $AD \cong AB$, and the angle at A right. I.47 then gives equality of the square on DC with the sum of the squares on DA and AC. The hypothesis gives the same sum as the square on BC, hence the squares on DC and BC are equal. SSS then compares ADC with ABC and transfers the right angle.

Wilkins' study note is particularly useful because it isolates exactly the chain

$$\mathrm{Sq}(DC) = \mathrm{Sq}(DA) + \mathrm{Sq}(AC) = \mathrm{Sq}(BA) + \mathrm{Sq}(AC) = \mathrm{Sq}(BC),$$

followed by the statement that DC and BC are equal and then by I.8. The formal proof preserves this outer chain.

The point at which the reconstruction expands the classical argument is the transition from equal squares to equal sides. The classical tradition itself recognized this as a separate converse statement. Heath, in his commentary on I.46, records Proclus' proof of both directions: squares on equal straight lines are equal, and conversely equal squares stand on equal straight lines. The current term-level scissors semantics therefore does not invent a new problem; it makes an old converse-content step explicit as a theorem.

10.2 Hartshorne: equal content and property (Z)

Hartshorne's discussion of Euclid's area theory is the most useful semantic control for the strictness issue. He distinguishes the equal-content relation from numerical area and lists the "whole is greater than the part" property as the condition that a properly contained figure cannot have the same content as the whole. This is property (Z) in his development.

For I.48 Hartshorne makes the dependence completely explicit: Euclid uses the fact that squares with equal content have equal sides, and Hartshorne states that this step depends on (Z); Exercise 22.6 asks precisely for the theorem that equal-content squares stand on congruent segments, assuming (Z). Thus the central local theorem of the Lean file,

```
i48_congruent_of_equal_squares
```

lands exactly on the semantic boundary identified in the modern synthetic analysis of Euclid.

That observation explains the formal architecture accurately. The equality operations of the scissors calculus support the forward Pythagorean theorem, but the converse step needs a strictness principle. The present I.48 proof derives the required square-specific (Z) consequence from the project-wide triangle positivity assumption and an explicit inner-square decomposition. It does not claim that triangle positivity and Hartshorne's full property (Z) are definitionally or logically identical in the current API.

10.3 Hilbert: Theorem 48 and De Zolt

Hilbert's Theorem 48 in Chapter IV should not be confused with Euclid's Proposition I.48. Hilbert's theorem is the converse of his equal-base, equal-altitude triangle theorem: equicomplementable triangles with the same base have the same altitude. Hilbert identifies it with Euclid I.39. It is therefore a general converse-content theorem rather than a statement about Pythagoras.

The structural relevance to the present proposition is nevertheless strong. Hilbert's later area theory shows how equal content can recover metric information, and his De Zolt theorem states that after a rectilinear figure is decomposed into triangular pieces, omitting a genuine triangle cannot leave enough content to reconstruct the whole.

The current Lean development does not import Hilbert's numerical segment-arithmetic proof of these results. Instead, it isolates the needed strictness in

```
hilbert_scissors_triangle_positive
```

and derives the square-specific proper-part theorem synthetically from finite triangle splits.

10.4 Borsuk–Szmielew and Greenberg

Borsuk–Szmielew and Greenberg remain control sources for the underlying synthetic geometry: incidence, order, congruence, segment construction, perpendiculars, parallelograms, and separation arguments. Proposition I.48 does not import a coordinate model or an ordered-field representation from those sources.

Their role is therefore geometric rather than semantic. The content strictness is kept explicitly in the Hilbert scissors layer instead of being hidden inside coordinates.

10.5 Geometry, content, and representation audit

For the required three-way audit:

- **Geometry**: moderate in the outer proof and substantial in the strictness core. The proof erects a prescribed perpendicular segment, constructs an inner square from strict segment comparison, establishes strict betweenness, and proves nondegeneracy of the remainder triangles.
- **Content**: central. The proof uses I.47, square transport, exact triangle splits, equicomplementability transitivity, cancellation, and the global positivity principle to derive strict monotonicity of square content.
- **Representation**: substantial but separable. Explicit square choices, alternative parallelogram triangulations, multiset regrouping, and conversion between a triangle list and a scissors term all appear.

11 Dependency Summary

The actual formal dependency chain is

$$\begin{gathered}
\text{Hilbert incidence + order + congruence + Group IV} \\
\Downarrow \\
\text{segment order / SameRay / right angles / parallelograms} \\
\Downarrow \\
\text{Hilbert scissors equality and equicomplementability} \\
\Downarrow \\
\text{I.42 + I.45 + I.46 + I.47 infrastructure} \\
\Downarrow \\
\texttt{i48_erect_perpendicular} + \texttt{i48_square_transport} \\
\Downarrow \\
\text{strict side comparison} \\
\Downarrow \\
\text{inner square construction} \\
\Downarrow \\
\text{exact outer = inner + four-triangle remainder split} \\
\Downarrow \\
\text{remainder nondegeneracy + I.45 reduction} \\
\Downarrow \\
\texttt{hilbert_scissors_triangle_positive} \\
\Downarrow \\
\text{square proper-part theorem} \\
\Downarrow \\
\texttt{i48_aux_square_not_equal_of_less} \\
\Downarrow \\
\text{Book Zero segment trichotomy} \\
\Downarrow \\
\texttt{i48_congruent_of_equal_squares} \\
\Downarrow \\
\text{I.47 on } ADC + \text{square transport} \\
\Downarrow \\
DC \cong BC \\
\Downarrow \\
\text{SSS + right-angle transport} \\
\Downarrow \\
\texttt{euclid_proposition_48}.
\end{gathered}$$

The contrast with the classical DAG is

$$\begin{gathered}
\text{Euclid: I.11 + I.3 + I.47 + equal squares have equal sides + I.8} \\
\Downarrow \\
\text{I.48,}
\end{gathered}$$

while the Lean proof expands the equal-squares step into a separate proper-part and positivity theory.

New geometric axiom in I.48:	no
New content axiom in I.48:	no
Inherited temporary content assumption:	yes, scissors positivity
New proposition-local helpers:	yes
New reusable local helper candidates:	yes
New Book Zero theorem:	no
New Interface/Scissors theorem:	no
Numerical area semantics:	no
Segment multiplication / field theory:	no
Exact scissors dissection used:	yes
Equicomplementability used:	yes
Hilbert Group IV available/used transitively:	yes
Continuity assumption in the file:	no
Archimedean assumption in the file:	no
Public theorem compiles:	yes
Full `lake build`:	clean
Active proposition-local `axiom`:	no
`sorry`/`admit` in `Proposition48.lean`:	no

12 Conclusion

Proposition I.48 closes Book I by reversing the Pythagorean theorem at the level at which Euclid actually states it: equality of plane content, not an equation between numerical squared lengths.

The outer proof remains close to Euclid. A prescribed perpendicular segment creates a right triangle ADC with $AD \cong AB$. Proposition I.47 identifies the square on DC with the sum of the two leg squares. Transport replaces those squares by the ones occurring in the hypothesis. Once $DC \cong BC$ is known, SSS transfers the right angle immediately.

The main formal contribution is proving that intermediate side congruence. The source does not assume that equal square content implies congruent sides. Instead it proves strict monotonicity: a strictly shorter side produces a strictly smaller square in the scissors sense. The proof constructs an inner square, exposes the outer square as that inner square plus an explicit four-triangle remainder, proves the remainder nondegenerate, reduces it to one nondegenerate triangle, and invokes the already isolated scissors positivity principle. Segment trichotomy then converts the exclusion of both strict orders into congruence.

This makes the foundational status transparent. No new I.48-specific axiom is introduced, and the two provisional local assumptions from the original sketch have both disappeared. The remaining debt is the project-wide configuration-free positivity principle already isolated in `HilbertScissorsPositivity`. No numerical area, segment multiplication, field construction, continuity axiom, or Archimedean axiom is added to close the proposition.

The final Book I theorem is therefore both synthetic and modular: the classical converse-Pythagoras shell is short, while the nontrivial converse content principle has been factored into a reusable strict-square argument whose exact dependence on scissors positivity is visible in the code.